SHAWN RAO

shawnrao@umich.edu | 734-272-8732 | shawnrao21.github.io | Ypsilanti, MI

SUMMARY

Data Science undergraduate in the University of Michigan College of Engineering, graduating December 2027. Works in C++ on performance-sensitive systems, in Python and R on data and statistics, and in JavaScript on accessible web interfaces. Seeking a Summer 2027 internship in software engineering or data.

EDUCATION

University of Michigan, College of Engineering Ann Arbor, MI
Bachelor of Science in Engineering (B.S.E.), Data Science *Expected 12/2027*

Relevant Coursework: EECS 281 (Data Structures & Algorithms), EECS 280, EECS 203 (Discrete Math), EECS 376 (Foundations of CS), SI 339 (Web Design & Accessibility), SI 206, ROB 101 (Computational Linear Algebra), MATH 215 (Vector Calculus), STATS 250

In Progress (Fall 2026): DATASCI 306 (Statistical Computing in R), STATS 412 (Probability & Statistics), EECS 201 (Unix, Shell, Version Control)

Purdue University *2025*
Applied Generative AI Specialization (online, 16 weeks)

TECHNICAL SKILLS

Languages: C++, Python, R, SQL, JavaScript (ES6+), HTML5, CSS3
Algorithms & Data Structures: Heaps and priority queues, union-find with path compression, graph algorithms (Prim, Dijkstra, MST), dynamic programming, branch and bound, hashing, BST and AVL trees, C++ STL
Statistics: Probability distributions (Binomial, Poisson, Exponential, Normal), joint and conditional distributions, transformations of random variables, Central Limit Theorem, point estimation, propagation of measurement error, OLS regression and confounding, t-tests
Data Tools: R, tidyverse, RStudio, Quarto, Pandas, NumPy, JMP, SQL (window functions, recursive CTEs)
Front End & APIs: DOM manipulation, CSS Grid, Flexbox, Media Queries, responsive design, WCAG 2.2 accessibility, React and Node.js (self-directed study), LangChain, OpenAI API
Testing & Verification: Randomized differential testing, mutation testing, Valgrind memory checking, strict-flag builds (`-Wall -Wextra -Werror`), WCAG contrast and keyboard audits
Version Control: Git (interactive rebase, hooks, reflog recovery, conflict resolution, remote tracking, stash), GitHub, GitLab, collaborative workflows
Systems & Shell: Linux, Unix, Bash, Zsh, shell scripting, process control (fork/exec/wait), grep, sed, POSIX regex, SSH key authentication, WSL, Docker, AWS EC2 and S3

PROJECTS

Order-Matching Engine and Market Simulator *C++, STL*

- Built a discrete-event engine that reads a stream of timestamped buy and sell orders and matches them against a price-time priority book, using paired binary heaps so each insert and match runs in $O(\log n)$.
- Produced running median price, per-trader transfer totals, and time-travel arbitrage reports in a single pass over the order stream, with no second read of the input.

Shell Tooling and Git Automation *C, Bash, Zsh, Git, sed*

- Implemented a Unix shell in C using fork, execvp and waitpid, with cd and exit handled in the parent because a child process cannot change its parent's working directory.
- Wrote a pre-commit Git hook that rejects commits containing SSH private keys or hard-coded API credentials, and a Zsh and Bash prompt reporting branch state, untracked files, divergence from the remote and last command runtime.

EXPERIENCE

University of Michigan, College of Engineering (CAEN) Ann Arbor, MI
Computer Hardware Technician, Student Staff *04/2023 – 04/2026*

- Supported 200+ Linux, Windows and macOS workstations and servers in shared engineering labs, holding 99%+ uptime for 500+ students and faculty.
- Reduced hardware downtime 25% by standardizing imaging and diagnostics and tracking recurring failure modes.
- Supported a CrowdStrike endpoint rollout to full fleet coverage; integrated 3 new lab technologies, cutting setup time 40%.

NYX Inc. Livonia, MI
Information Technology Summer Intern *04/2021 – 07/2021*